A universal numerical renormalization method for the combined Duffing–Van der Pol oscillator

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We present a single, *universal* numerical renormalization engine, based on the dynamical renormalization group (DRG), that constructs accurate approximate solutions of the harmonic oscillator deformed simultaneously by a Duffing cubic stiffness and a Van der Pol nonlinear damping. Working only with the first-order perturbative response, the method extracts amplitude- and phase-flow generators by projecting the secular growth of the perturbation onto the tangent directions of the background solution. Because the first-order problem is linear, the Duffing and Van der Pol generators superpose, so the same engine reproduces the pure Duffing frequency shift $\beta_\theta = \frac{3}{8}A^2$ (recovering the Bender–Orszag multiple-scales result), the pure Van der Pol amplitude flow $\beta_A = \frac{1}{2}A(1-A^2/4)$ with its limit cycle at $A = 2$, and the combined dynamics—for arbitrary deformation strengths and free of secular divergences. We benchmark the integrators used for the reference solution and discuss the method's accuracy from the weak- to the strong-coupling regime.

I. INTRODUCTION

Control system engineering integrates efficient process design with technical forecasting and underpins applications in manufacturing, transportation, and environmental management [1]. A central difficulty is selecting a mathematical framework that captures how a system responds to internal and external fluctuations. Differential equations provide such a framework, but exact solutions are rarely attainable, and perturbation theory—while powerful—often develops singular (secular) behaviour that grows without bound in time.

One technique that tames secular growth is the dynamical renormalization group (DRG) [2, 3], which reuses the first-order perturbative solution to reparameterize the background, capturing the nonlinear features of the dynamics as the parameters flow in time. Obtaining closed-form perturbative solutions is not always possible, so the DRG has been adapted to accept both analytic and numerical input [4]. In this work we build a *numerical* DRG that, from the first-order response alone, produces solutions of the Duffing and Van der Pol equations as deformations of the simple harmonic oscillator, and—our main contribution—of the two deformations acting *together*, through a single universal engine.

Section II introduces the method and its numerical implementation, reproducing the Duffing (frequency-shift) and Van der Pol (amplitude-flow) cases. Section III develops the combined Duffing–Van der Pol (DVdP) problem and shows how the independent generators superpose. Section IV benchmarks the numerical integrators and the accuracy of the renormalized solution across coupling strengths. Section V discusses case studies relevant to nonlinear control, and Section VI concludes.

II. METHOD OF NUMERICAL RENORMALIZATION

A. Deforming the harmonic oscillator

We start from the harmonic oscillator written as a first-order system,

$$\frac{d\varphi^{(0)}}{dt} = \Pi^{(0)}, \qquad \frac{d\Pi^{(0)}}{dt} = -\omega^2\varphi^{(0)}, \tag{1}$$

and deform it into the damping- and force-free Duffing equation,

$$\frac{d\varphi_{\text{full}}}{dt} = \Pi_{\text{full}}, \tag{2a}$$

$$\frac{d\Pi_{\text{full}}}{dt} = -\omega^2\varphi_{\text{full}} - \epsilon\,\varphi_{\text{full}}^3, \tag{2b}$$

by a cubic correction controlled by the tunable parameter ϵ. With maximum initial amplitude set to one,

$$\varphi^{(0)}(t) = \cos\varphi_\circ, \qquad \Pi^{(0)}(t) = -\omega\sin\varphi_\circ, \tag{3}$$

where $\varphi_\circ = \omega t$ is the oscillation phase, obeying $d\varphi_\circ/dt = \omega$.

Since Eq. (2b) is nonlinear, we seek small corrections to the background,

$$\varphi_{\text{full}} \approx \varphi^{(0)} + \epsilon\varphi^{(1)}, \qquad \Pi_{\text{full}} \approx \Pi^{(0)} + \epsilon\Pi^{(1)}, \tag{4}$$

Substituting into Eq. (2b) and collecting the terms of order ϵ^1 gives the driven linear system

$$\frac{d\varphi^{(1)}}{dt} = \Pi^{(1)}, \tag{5a}$$

$$\frac{d\Pi^{(1)}}{dt} = -\omega^2\varphi^{(1)} - \left(\varphi^{(0)}\right)^3. \tag{5b}$$

The source $(\varphi^{(0)})^3$ acts as a resonant forcing; integrating with $\varphi^{(1)} = \Pi^{(1)} = 0$ produces an amplitude that grows after a few oscillations. This *secular growth* (Fig. 1) is a well-known artifact of the perturbative decomposition [5, 6], not of the full solution.

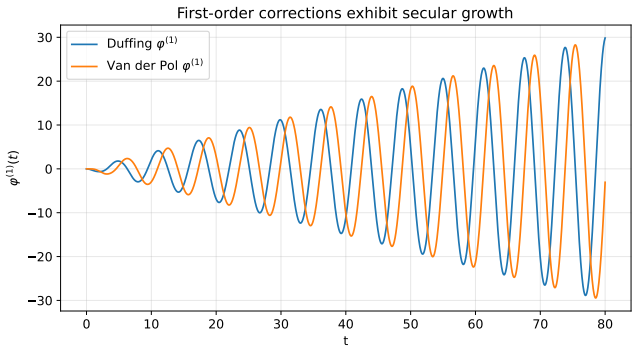


FIG. 1. First-order corrections $\varphi^{(1)}$ for the Duffing and Van der Pol deformations about $A = 1$. Both grow without bound in time, so the naive sum $\varphi^{(0)} + \epsilon\varphi^{(1)}$ fails at long times.

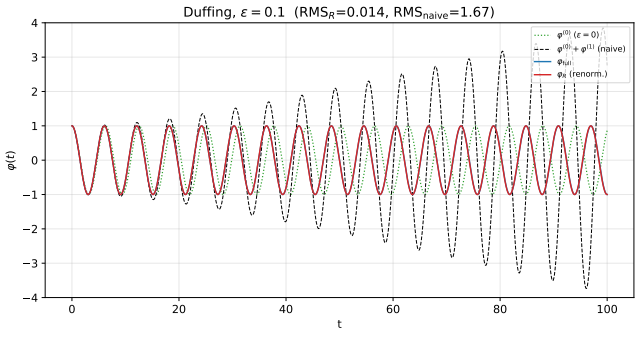


FIG. 2. Duffing oscillator at $\epsilon = 0.1$: the full solution $\varphi_{\rm full}$ (blue), the renormalized solution φ_R (red) and the naive first-order solution $\varphi^{(0)} + \epsilon\varphi^{(1)}$ (dashed). The renormalized solution captures the phase evolution without secular amplitude growth. The RMS errors versus the full solution are quoted in the title.

B. Renormalized solutions from first-order perturbation theory

The full Duffing solution differs from the cosine mainly in its oscillation phase, suggesting that a time-dependent correction to the frequency and phase suffices. We promote the background parameters $\vec{\lambda}_0 = (\varphi_\circ; \omega)$ to flow in time through the infinitesimal transformation

$$\vec{\lambda}_0(t) \to \vec{\lambda}_R(t) = \vec{\lambda}_0(t) + \epsilon\,\Delta\vec{\lambda}, \tag{6}$$

which induces the parameter-flow equations

$$\frac{d\varphi_\circ^R}{dt} = \omega_R + \epsilon\,\alpha_\omega^{(1)}, \qquad \frac{d\omega_R}{dt} = \epsilon\,\beta_\omega^{(1)}. \tag{7}$$

The renormalized solution is obtained by inserting the flowing parameters back into the background,

$$\varphi_{\rm full} \approx \varphi_R(t) \equiv \varphi^{(0)}(t, \vec{\lambda}_R), \qquad \Pi_{\rm full} \approx \Pi_R(t). \tag{8}$$

The generators are fixed by matching the first-order expansion to the expanded renormalized solution. Writing $\varphi_R = \varphi^{(0)} + \epsilon\,\nabla_{\vec{\lambda}_0}\varphi^{(0)} \cdot \Delta\vec{\lambda}$ and collecting the $O(\epsilon)$ terms yields the residual

$$\varphi_{\perp,A}^{(1)} = \varphi_A^{(1)} - \varphi_{\parallel,A}, \qquad \varphi_{\parallel,A} \equiv \nabla_{\vec{\lambda}_0}\varphi_A^{(0)} \cdot \Delta\vec{\lambda}. \tag{9}$$

Minimizing the cost function

$$I = \sum_{A=1}^{2} \int_{t_0}^{\infty} \left|\varphi_{\perp,A}^{(1)}\right|^2 dt \tag{10}$$

interprets the generators as the projection coefficients of the perturbation $\varphi_A^{(1)}$ onto the basis of background derivatives $\nabla_{\vec{\lambda}}\varphi_A^{(0)}$ (Fig. 4). Because the cost is quadratic, the minimization reduces to a small linear system.

a. Numerical generator extraction. In practice we parameterize the background by an amplitude A and a phase θ, $\varphi^{(0)} = A\cos(\omega t + \theta)$, and decompose the numerically integrated perturbation $(\varphi^{(1)}, \Pi^{(1)})$ *pointwise* onto the tangent directions $\partial_A\varphi^{(0)}$ and $\partial_\theta\varphi^{(0)}$ (and their momentum partners). A resonant generator makes the resulting coefficient time series grow linearly; its slope is the generator. This isolates the secular content from bounded oscillatory contributions and is numerically robust up to strong coupling. For the Duffing oscillator we find that the amplitude generator vanishes, $\beta_A \approx 0$, while the phase generator is

$$\beta_\theta^{\rm Duf}(A) = \tfrac{3}{8}A^2, \tag{11}$$

in exact agreement with the multiple-scales result of Bender and Orszag [5] at $A = 1$, $\omega = 1$. Solving the flow $d\theta/dt = \epsilon\,\beta_\theta(A)$ and reconstructing φ_R gives a solution that stays in phase with the full Duffing solution for tens of oscillations and never diverges in amplitude (Fig. 2).

C. The Van der Pol oscillator

The Van der Pol oscillator is a different deformation of the harmonic oscillator,

$$\frac{d\varphi}{dt} = \Pi, \qquad \frac{d\Pi}{dt} = -\omega^2\varphi + \alpha\,(1 - \varphi^2)\,\Pi, \tag{12}$$

with strength α. (We use the standard sign $1 - \varphi^2$ of the nonlinear damping.) Here the full solution spirals toward a limit cycle, so rather than a frequency shift the background must be parameterized by a *varying amplitude*, $\varphi^{(0)} = A_0\cos(\omega t)$, $\Pi^{(0)} = -\omega A_0\sin(\omega t)$. The first-order perturbation

$$\frac{d\Pi^{(1)}}{dt} = -\omega^2\varphi^{(1)} + \Pi^{(0)}\left(1 - (\varphi^{(0)})^2\right) \tag{13}$$

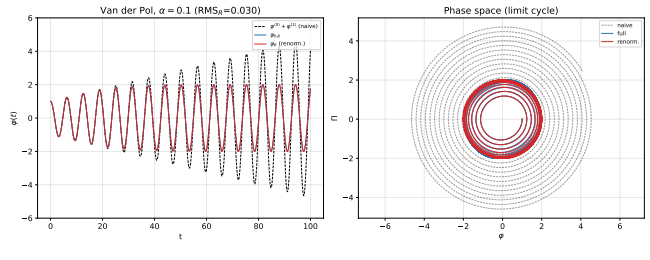


FIG. 3. Van der Pol oscillator at $\alpha = 0.1$. Left: time series of the full, renormalized and naive solutions. Right: phase-space portrait; the renormalized amplitude flow reproduces the approach to the limit cycle.

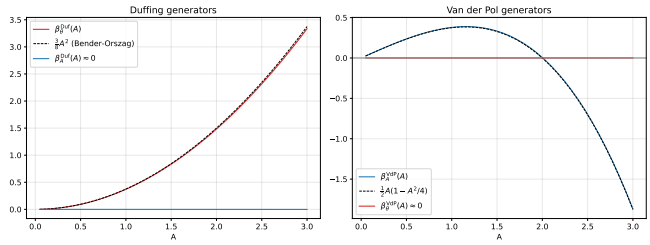


FIG. 4. Numerically extracted generators. Left: Duffing, where the phase generator follows $\frac{3}{8}A^2$ and the amplitude generator is consistent with zero. Right: Van der Pol, where the amplitude generator follows $\frac{1}{2}A(1 - A^2/4)$ (zero at the limit cycle $A = 2$) and the phase generator vanishes.

again grows secularly; projecting it onto the amplitude direction gives the amplitude flow

$$\frac{dA_R}{dt} = \alpha\,\beta_A(A_R), \qquad \beta_A^{\mathrm{VdP}}(A) = \tfrac{1}{2}A\big(1 - \tfrac{A^2}{4}\big), \quad (14)$$

whose stable zero at $A = 2$ is the limit-cycle radius. The reconstructed solution tracks the amplitude growth of the full Van der Pol solution and converges to the correct limit cycle without secular growth (Fig. 3).

III. THE COMBINED DUFFING–VAN DER POL OSCILLATOR

We now switch on both deformations simultaneously, with *independent* strengths ϵ (Duffing) and α (Van der Pol):

$$\frac{d\varphi}{dt} = \Pi, \qquad \frac{d\Pi}{dt} = -\omega^2\varphi - \epsilon\,\varphi^3 + \alpha\,(1 - \varphi^2)\,\Pi. \quad (15)$$

Setting $\epsilon = 0$ recovers Van der Pol and $\alpha = 0$ recovers Duffing, so Eq. (15) interpolates continuously between the two.

The key observation is that the first-order perturbation equation is *linear* in $(\varphi^{(1)}, \Pi^{(1)})$, with a source that is the sum of the Duffing and Van der Pol contributions evaluated on the *same* background. Superposition therefore holds,

$$\varphi^{(1)} = \epsilon\,\varphi^{(1)}_{\mathrm{Duf}} + \alpha\,\varphi^{(1)}_{\mathrm{VdP}}, \quad (16)$$

and the generators of the combined flow are the linear combination of the *independently* extracted generators,

$$\beta_A(A) = \epsilon\,\beta_A^{\mathrm{Duf}}(A) + \alpha\,\beta_A^{\mathrm{VdP}}(A), \quad (17)$$

$$\beta_\theta(A) = \epsilon\,\beta_\theta^{\mathrm{Duf}}(A) + \alpha\,\beta_\theta^{\mathrm{VdP}}(A). \quad (18)$$

This is what makes the method *universal*: a single engine extracts β^{Duf} and β^{VdP} once, and any combination (ϵ, α) is obtained by Eq. (18) followed by integration of the two-dimensional parameter flow

$$\frac{dA}{dt} = \beta_A(A), \qquad \frac{d\theta}{dt} = \beta_\theta(A), \quad (19)$$

and reconstruction $\varphi_R(t) = A(t)\cos\big(\omega t + \theta(t)\big)$. The Duffing sector supplies the (amplitude-dependent) frequency shift while the Van der Pol sector drives the amplitude toward the limit cycle; both act at once. Crucially, integrating Eq. (19) as a full nonlinear ODE—rather than truncating to a linear-in-Δt shift—keeps the construction valid well beyond the infinitesimal regime.

Figure 5 shows the combined solution for soft ($\epsilon = \alpha = 0.1$) and strong ($\epsilon = \alpha = 1$) coupling. For soft coupling the renormalized solution is nearly indistinguishable from the full one. For strong coupling it still captures the amplitude envelope and the size of the limit cycle, with a residual phase drift at late times: the single-harmonic ansatz $A\cos(\omega t + \theta)$ cannot represent the anharmonic shape that the true strong-coupling limit cycle develops. This is the expected limitation of a first-order scheme, and it is reached only deep in the strong-coupling regime.

IV. BENCHMARKING

A. Choice of integrator

The renormalization method only requires the first-order response, but a reference ("full") solution is needed to assess accuracy. We compared a fixed-step fourth-order Runge–Kutta against the adaptive Runge–Kutta (`RK45`), Dormand–Prince (`DOP853`) and implicit (`Radau`) integrators of SciPy [7] on the Duffing, Van der Pol and combined systems. `DOP853` delivered the best accuracy per right-hand-side evaluation on the non-stiff cases and reproduced the Van der Pol limit cycle, while `Radau` was the most accurate but an order of magnitude more expensive. We adopt `DOP853` (relative tolerance 10^{-10}) as the default reference integrator; the full table is provided in the repository.

B. Accuracy across coupling strengths

Figure 6 reports the root-mean-square error of the renormalized and naive solutions, relative to the full so-

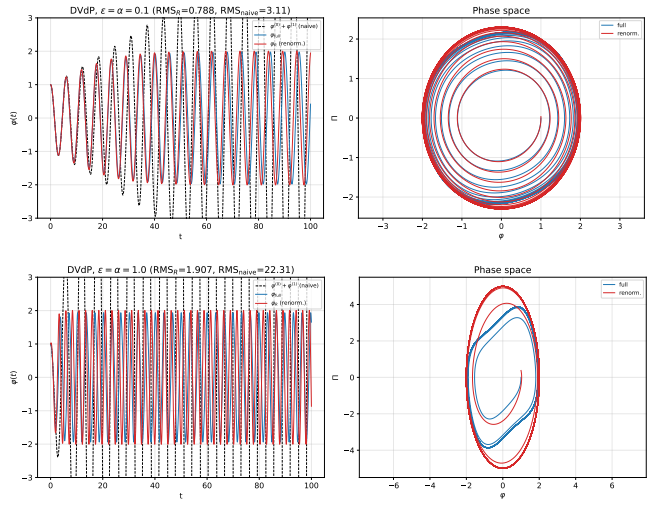


FIG. 5. Combined DVdP oscillator. Top: soft coupling $\epsilon = \alpha = 0.1$. Bottom: strong coupling $\epsilon = \alpha = 1$. Left panels: time series (full, renormalized, naive). Right panels: phase-space portraits. The renormalized solution tracks the full one closely at soft coupling and captures the envelope and limit-cycle size at strong coupling.

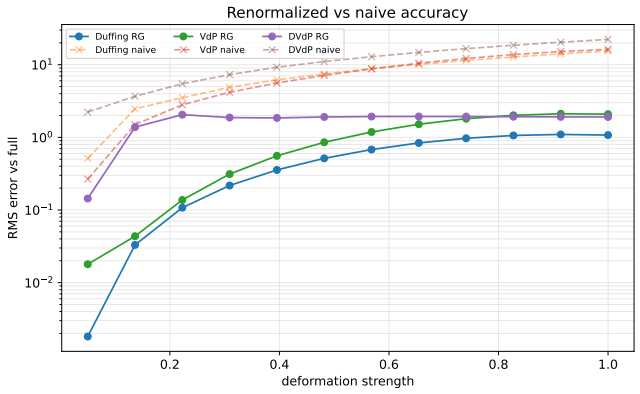


FIG. 6. RMS error versus the full solution as a function of deformation strength, for Duffing, Van der Pol and the combined system. Solid: renormalized; dashed: naive first-order. The renormalized solution is consistently more accurate and avoids secular divergence.

lution over $t \in [0, 100]$, as a function of the deformation strength. The renormalized solution outperforms the naive perturbative one by roughly an order of magnitude throughout, for all three systems, and remains bounded where the naive solution diverges secularly.

V. CASES OF STUDY

The combined DVdP oscillator is a minimal model of a self-sustained nonlinear actuator with a stiffening restoring force, a setting common in nonlinear control. Two benchmarking directions follow naturally from the method:

1. *Semi-analytical surrogate.* Because the generators are extracted once and stored as functions of amplitude, the renormalized flow Eq. (19) provides a fast, low-dimensional surrogate for the oscillator's envelope and phase, suitable for real-time estimation where only the linear (first-order) response of the plant is available.

2. *Limit-cycle characterization.* The amplitude flow directly yields the limit-cycle radius and its dependence on (ϵ, α), which can be compared against the Hopf-bifurcation analysis of related van der Pol–Mathieu–Duffing models [8].

A systematic comparison against multiple-scale and harmonic-balance approximations, and an extension of the ansatz to include higher harmonics for the strong-coupling limit cycle, are left for future work.

VI. DISCUSSION

We have shown that a single numerical renormalization engine, built from the first-order perturbative response, reproduces the Duffing frequency shift, the Van der Pol amplitude flow, and—through the linear superposition of their generators—the combined Duffing–Van der Pol dynamics for independent and arbitrary deformation strengths, free of secular divergences. The method is accurate from the weak-coupling regime up to moderately strong coupling; its only essential limitation, a late-time phase drift at very strong coupling, is inherited from the single-harmonic background ansatz and can be addressed by enriching the parameter set. The construction requires only the linear response of the system, making it attractive for nonlinear control applications where a full analytic model is unavailable.

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